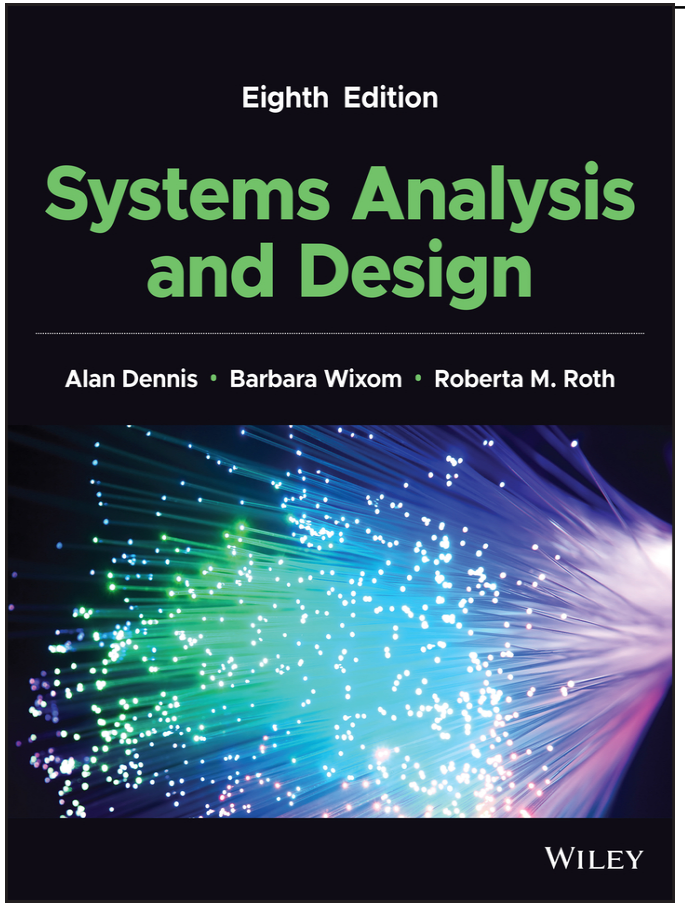
Software Development Lifecycles

Unit 3



Systems Analysis and Design

Chapter 8

User Interface Design

Objectives

- The Usability Concept
- Principles of user interface design.
- User interface design process.
- Navigation design.
- Input design.
- Output design.

Introduction

- **Interface design** is the process of defining how the system will interact with external entities.
- Focus on the design of ***user interfaces*** – how the system will interact with the users.
- The design of system interfaces defines how the systems exchange information with other systems.

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- The user interface includes three fundamental parts:

- The ***Navigation mechanism*** - the way in which the user tells the system what to do.

- The ***input mechanism*** – the way in which the system captures information.

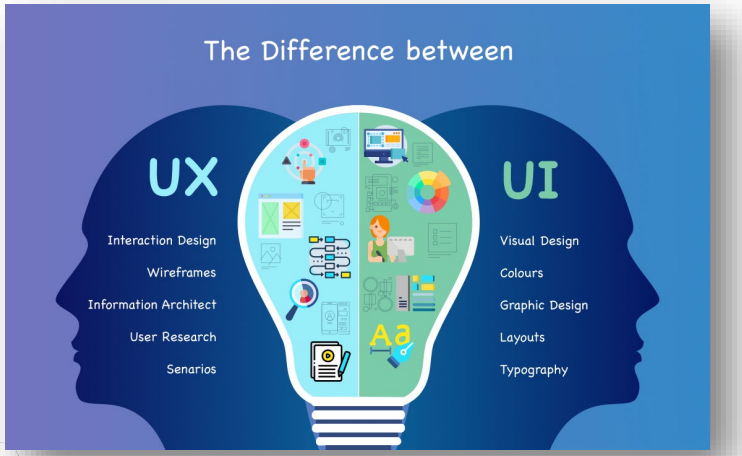
- The **output mechanism** - the way in which the system provides information to the user or to other systems.

- **Graphical user interfaces (GUI)** use windows, menus, icons, etc., and are the most common type of user interfaces.

The Usability Concept

- Users want an interface that is high on usability so that needed tasks are performed quickly and effectively.

- The term usability encompasses two related concepts: systems that are easy to use and easy to learn



Principles for User Interface Design

- User interface design is an art.
- The goal is to make the interface pleasing to the eye and simple to use, while minimizing the user's effort.

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Principle	Description
Layout	The interface should be a series of areas on the screen that are used consistently for different purposes—for example, a top area for commands and navigation, a middle area for information to be input or output, and a bottom area for status information.
Content awareness	Users should always be aware of where they are in the system and what information is being displayed.
Aesthetics	Interfaces should be functional and inviting to users through careful use of white space, colors, and fonts. There is often a trade-off between including enough white space to make the interface look pleasing and losing so much space that important information does not fit on the screen.
Usage level	Although ease of use and ease of learning often lead to similar design decisions, there is sometimes a trade-off between the two. Infrequent users of software will prefer ease of learning, whereas frequent users will prefer ease of use.
Consistency	Consistency in interface design enables users to predict what will happen before they perform a function. It is one of the most important elements in ease of learning, ease of use, and aesthetics.
Minimize user effort	The interface should be simple to use. Most designers plan on having no more than three mouse clicks from the starting menu until users perform work.

FIGURE 8-1 Principles of user interface design.

Layout

- ***Layout*** refers to organizing areas of the screen and document for different purposes and using these areas consistently throughout the user interface.
- The screen is often divided into three areas:
  - The top area provides the user with ways to navigate through the system;
  - The middle and the largest area is for display of user's work; and
  - The bottom area contains status information about that the user is doing.

(cont'd)

- The areas and information within areas should have a natural intuitive flow to minimize user's movement from one area to the next.
- Ideally, the areas will remain consistent in
  - size,
  - shape,
  - placement for the forms,
  - reports used to present it.

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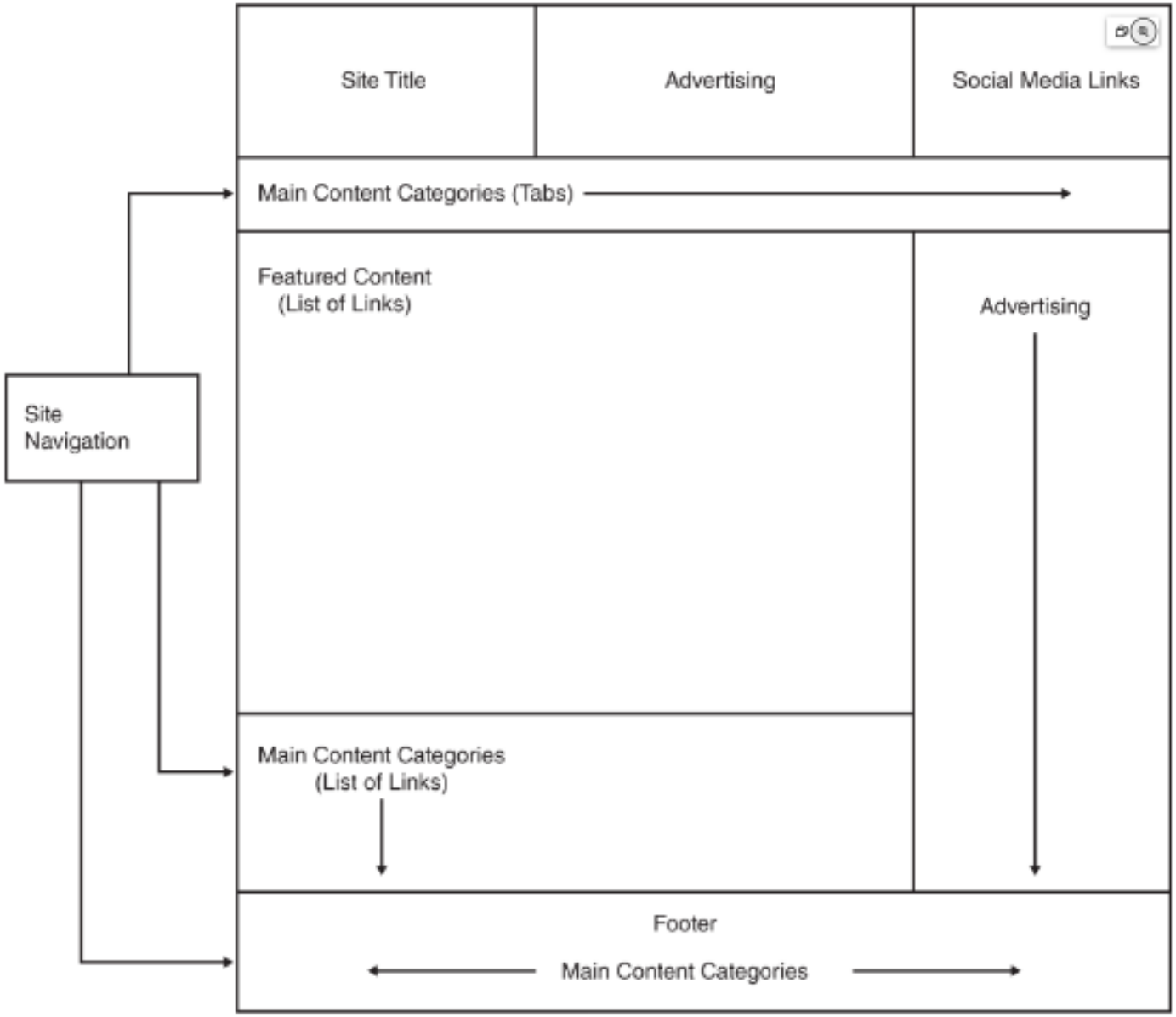
- Web page layout with multiple navigation areas



Content Awareness

- ***Content awareness*** refers to the ability of an interface to make the user aware of the information it contains.
- All interfaces should have titles.
- Menus should show where the user are and where the user came from to get there.
- All area should be clear and well defined.
- Content awareness also applies to the *fields* and *field labels* within each area, and the information that a form or report contains.

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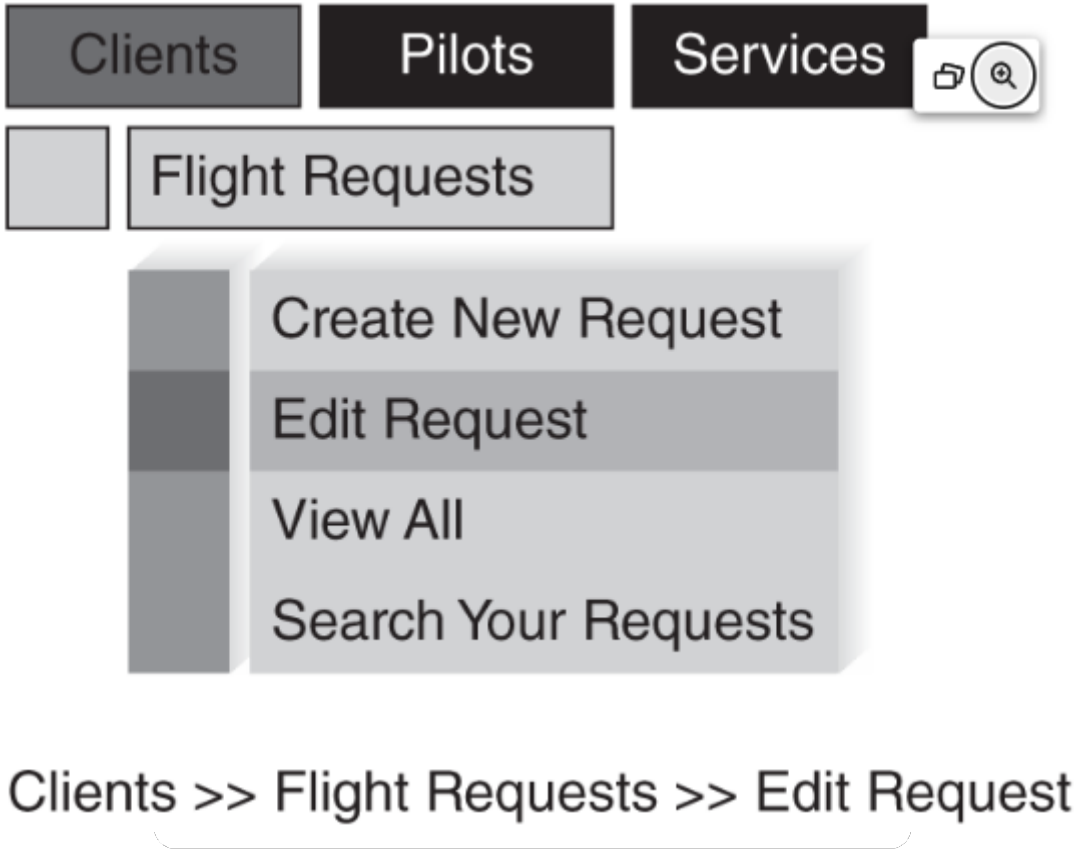
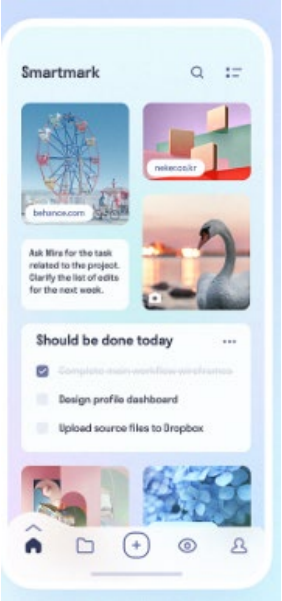
Content awareness with menu selection highlights and breadcrumbs.



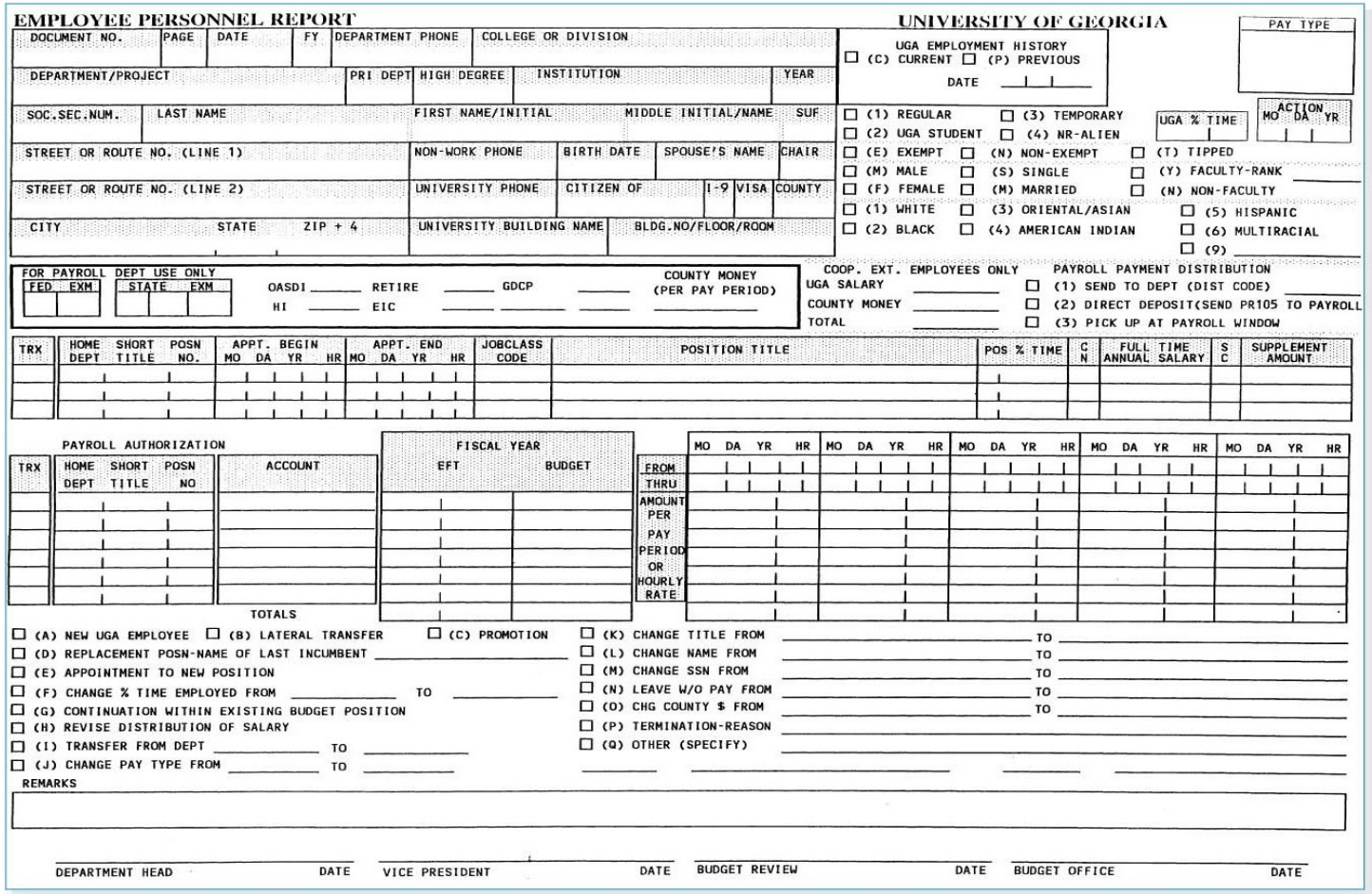
Aesthetics

- ***Aesthetics*** refers to designing interfaces that pleasing to the eye.
- Interfaces need to be functional and inviting to use.
- In general, all forms and reports need a certain amount of white space.
- The design of text is also important.
  - Fonts and font sizes
  - Colors and patterns



Usage Level

An example of form with high ***density***



EMPLOYEE PERSONNEL REPORT

UNIVERSITY OF GEORGIA

DOCUMENT NO. | PAGE | DATE | FY | DEPARTMENT PHONE | COLLEGE OR DIVISION

DEPARTMENT/PROJECT | PRI DEPT | HIGH DEGREE | INSTITUTION | YEAR

SOC.SEC.NUM. | LAST NAME | FIRST NAME/INITIAL | MIDDLE INITIAL/NAME | SUF

STREET OR ROUTE NO. (LINE 1) | NON-WORK PHONE | BIRTH DATE | SPOUSE'S NAME | CHAIR

STREET OR ROUTE NO. (LINE 2) | UNIVERSITY PHONE | CITIZEN OF | I-9 | VISA | COUNTY

CITY | STATE | ZIP + 4 | UNIVERSITY BUILDING NAME | BLDG.NO/FLOOR/ROOM

UGA EMPLOYMENT HISTORY
☐ (C) CURRENT ☐ (P) PREVIOUS
DATE

PAY TYPE

ACTION MO DA YR

UGA % TIME

☐ (1) REGULAR ☐ (3) TEMPORARY
☐ (2) UGA STUDENT ☐ (4) NR-ALIEN
☐ (E) EXEMPT ☐ (N) NON-EXEMPT ☐ (T) TIPPED
☐ (M) MALE ☐ (S) SINGLE ☐ (Y) FACULTY-RANK
☐ (F) FEMALE ☐ (M) MARRIED ☐ (N) NON-FACULTY
☐ (1) WHITE ☐ (3) ORIENTAL/ASIAN ☐ (5) HISPANIC
☐ (2) BLACK ☐ (4) AMERICAN INDIAN ☐ (6) MULTIRACIAL
☐ (9)

FOR PAYROLL DEPT USE ONLY
FED EXM | STATE EXM
OASDI RETIRE GDCP COUNTY MONEY (PER PAY PERIOD)
HI EIC

COOP. EXT. EMPLOYEES ONLY
UGA SALARY
COUNTY MONEY
TOTAL

PAYROLL PAYMENT DISTRIBUTION
☐ (1) SEND TO DEPT (DIST CODE)
☐ (2) DIRECT DEPOSIT(SEND PR105 TO PAYROLL
☐ (3) PICK UP AT PAYROLL WINDOW

TRX	HOME DEPT	SHORT TITLE	POSN NO.	APPT. BEGIN MO DA YR HR	APPT. END MO DA YR HR	JOBCLASS CODE	POSITION TITLE	POS % TIME	C N	FULL TIME ANNUAL SALARY	S C	SUPPLEMENT AMOUNT

PAYROLL AUTHORIZATION

TRX	HOME DEPT	SHORT TITLE	POSN NO	ACCOUNT	FISCAL YEAR EFT	FISCAL YEAR BUDGET
				TOTALS		

	MO DA YR HR	MO DA YR HR	MO DA YR HR	MO DA YR HR	MO DA YR HR
FROM					
THRU					
AMOUNT PER PAY PERIOD OR HOURLY RATE					

☐ (A) NEW UGA EMPLOYEE ☐ (B) LATERAL TRANSFER ☐ (C) PROMOTION
☐ (D) REPLACEMENT POSN-NAME OF LAST INCUMBENT
☐ (E) APPOINTMENT TO NEW POSITION
☐ (F) CHANGE % TIME EMPLOYED FROM TO
☐ (G) CONTINUATION WITHIN EXISTING BUDGET POSITION
☐ (H) REVISE DISTRIBUTION OF SALARY
☐ (I) TRANSFER FROM DEPT TO
☐ (J) CHANGE PAY TYPE FROM TO

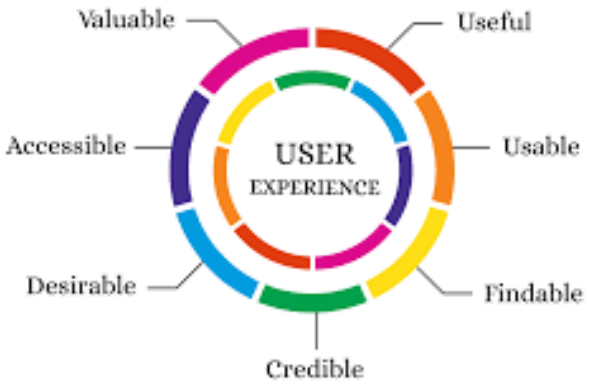
☐ (K) CHANGE TITLE FROM TO
☐ (L) CHANGE NAME FROM TO
☐ (M) CHANGE SSN FROM TO
☐ (N) LEAVE W/O PAY FROM TO
☐ (O) CHG COUNTY $ FROM TO
☐ (P) TERMINATION-REASON
☐ (Q) OTHER (SPECIFY)

REMARKS

DEPARTMENT HEAD DATE | VICE PRESIDENT DATE | BUDGET REVIEW DATE | BUDGET OFFICE DATE

User Experience

- ***User experience*** refers to designing the user interface with the users' level of computer experience in mind.
- Novice users are concerned with *easy of learning*.
- Expert users are concerned with *easy of use*.
- Often, the two objectives are complementary and lead to similar design decisions, but sometimes there are trade-offs.



Consistency

- ***Consistency*** usually refers to the interface within one computer system, so that all parts of the same system work in the same way. Ideally, however, the system also should be consistent with other computer systems in the organization.
- Consistency enables users to predict what will happen, and to reduce the amount of learning.
- Consistency occurs at many different levels.
  - Navigation controls;
  - Terminology;
  - Report and form design.

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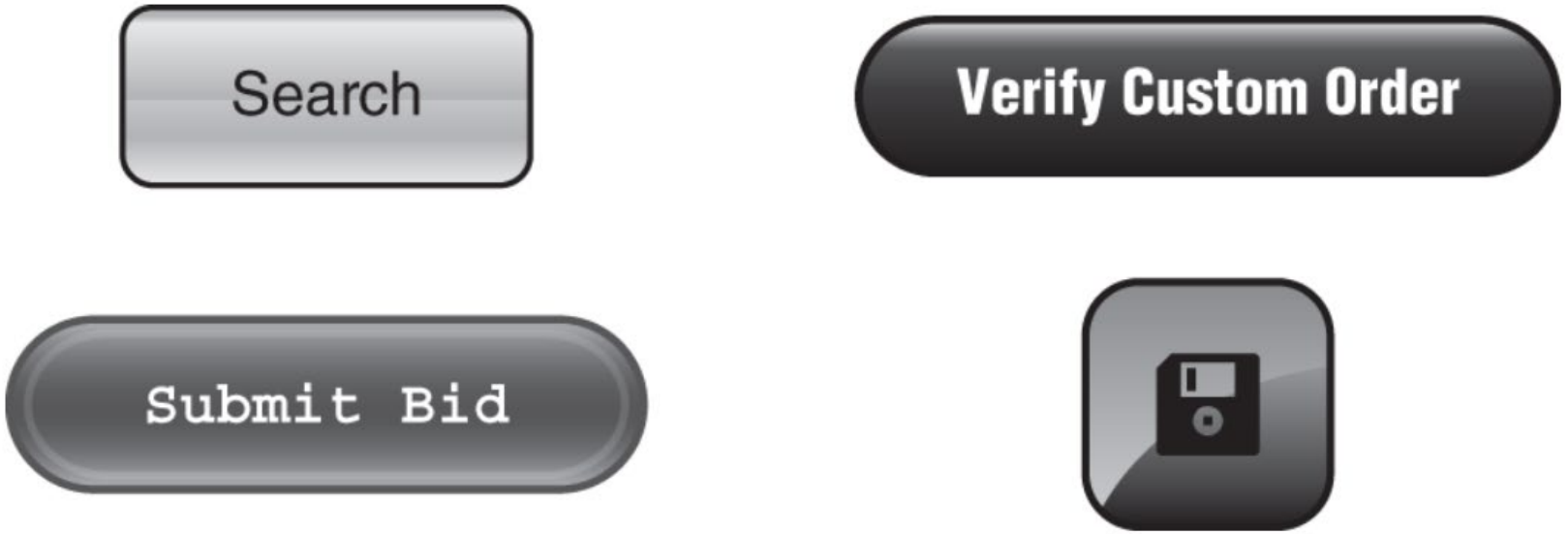

FIGURE 8-5 Inconsistent symbols and terminology.

Minimize User Effort

- ***Minimizing user effort*** means using the fewest possible mouse clicks or keystrokes to move from one part of the system to another.
- ***Three-clicks rule***
  - Users should be able to go from the start or main menu of a system to the information or action they want in no more than three mouse clicks or three keystrokes.

User Interface Design Process

- **User interface design** is a five-step process that is iterative.
- Analysts often move back and forth between components rather than proceed sequentially

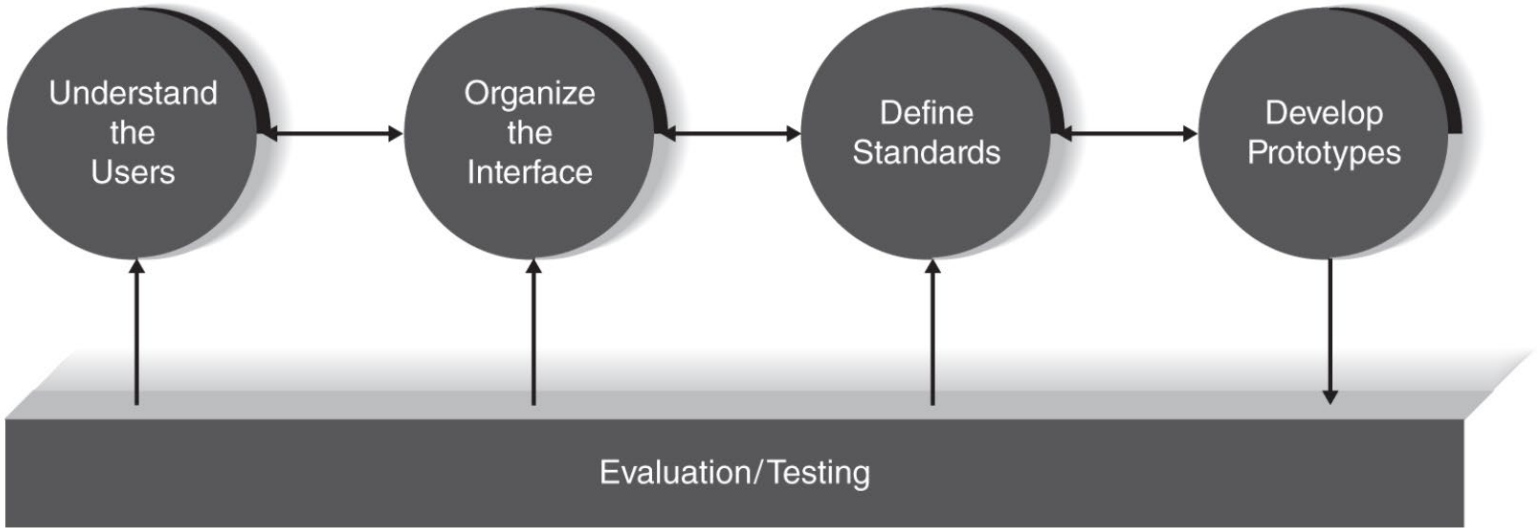

FIGURE 8-7 Interface design process.

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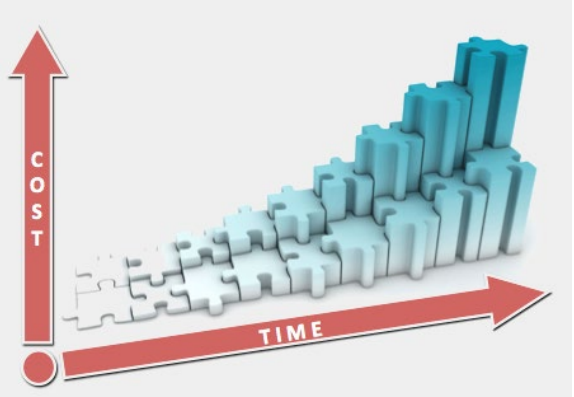
- In practice, most analysts interact closely with the users during the interface design process, so that users have many chances to see the interface as it evolves.
- The best advice is "test early; test often" when it comes to the user interface



Understand the Users

- Many websites and mobile applications targeting an array of external user audiences.
- Personas can be used to create descriptions of a fictional person who aggregates the common characteristics of a particular user group.
- By defining this person in terms of interests, typical behaviors, goals and objectives, and expectations, the analyst can plan for a user interface that will be satisfying for that user group

Use Scenario Development

- A ***use scenario*** is an outline of steps that user's perform to accomplish some part of their work.
- Use scenarios are presented in a simple narrative description that is tied to the DFD.

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Use Scenario: Focused Visitor	Use Scenario: Exploratory Visitor
User wants to easily develop alternative drone configurations and compare price and performance of the alternatives	User wants to learn about applications of commercial drones through video clips and written material
1. User selects a drone base model 2. User looks at lists of available options for that model 3. User selects options and wants to learn price and capabilities of that configuration 4. User saves, modifies, or discards that configuration and repeats with other options or other drones 5. User has one or more configurations to save and consider for purchase. 6. User may want to contact a sales representative with questions.	1. User views video clips of drones being used in commercial applications 2. User selects an application area and views more clips in that area 3. User reads site marketing material 4. User reads owner descriptions and comments 5. User repeats steps 2-4 for other commercial application areas. 6. User views list of drone models 7. User selects a model and reads site marketing material for that drone.

FIGURE 8-9 Two use scenarios for the DrōnTeq sales website.

Organize the Interface

- The ***interface structure design*** defines the basic components of the interface and how they work together to provide functionality to users.
- An interface structure diagram (ISD) is used to show how all screens, forms, and reports are related and how the user moves from one to another.
- An ISD is similar to a DFD in that it uses boxes and lines to show the structure. However, unlike DFDs, there are no commonly used rules or standards for ISDs.
- The basic structure of the interface follows the basic structure of the business process itself as defined in the process model.

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Interface Structure Diagram Example

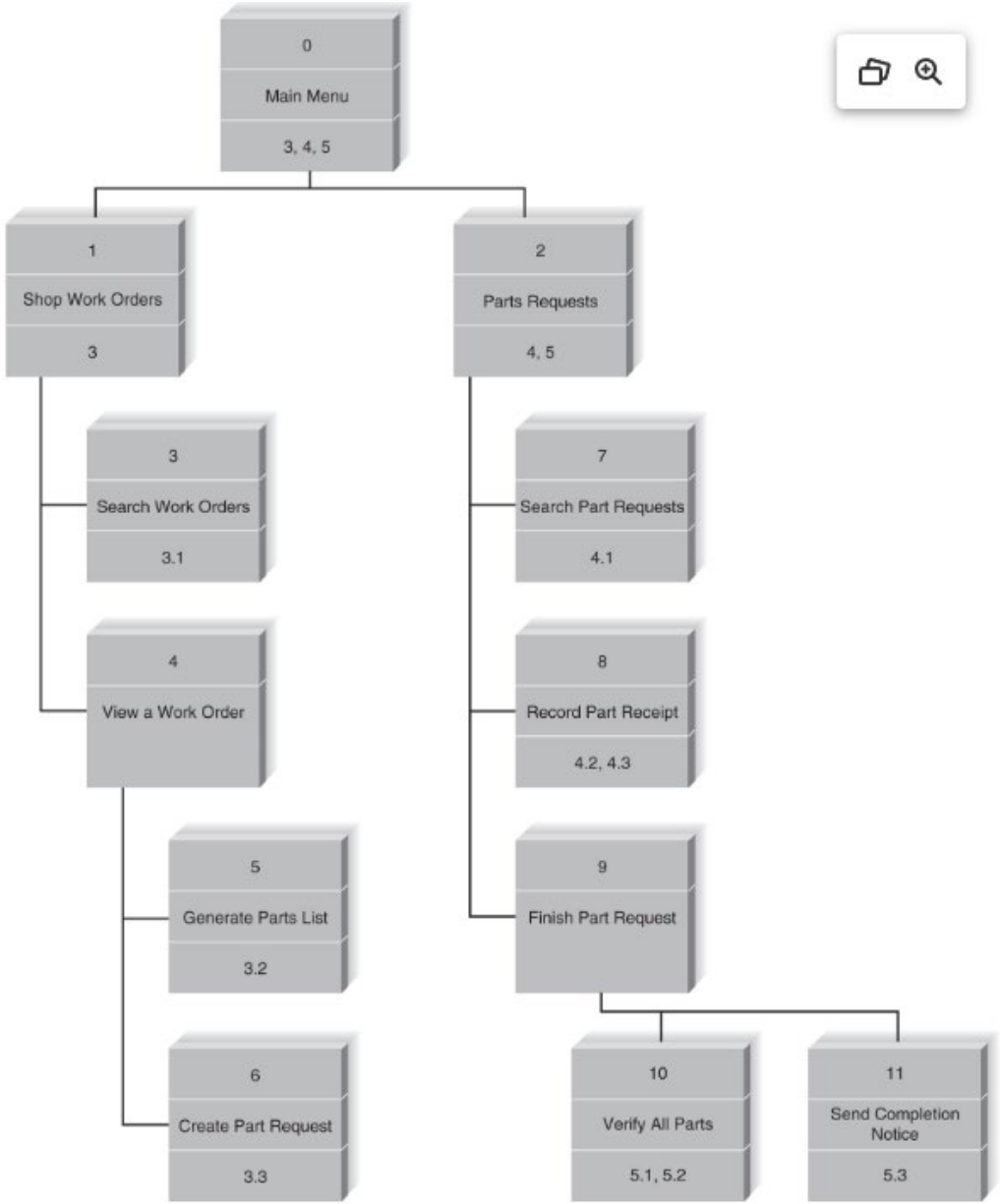

FIGURE 8-10 Shop management interface structure diagram.

Define Standards

- The interface standards are the basic design elements that are common across the individual screens, forms, and reports within the system.
- An ***interface metaphor*** is a concept from the real world that is used as a model for the computer system.
  - E.g., Quicken uses a checkbook metaphor.
- The ***interface template*** defines the general appearance of all screens and the paper-based forms and reports.

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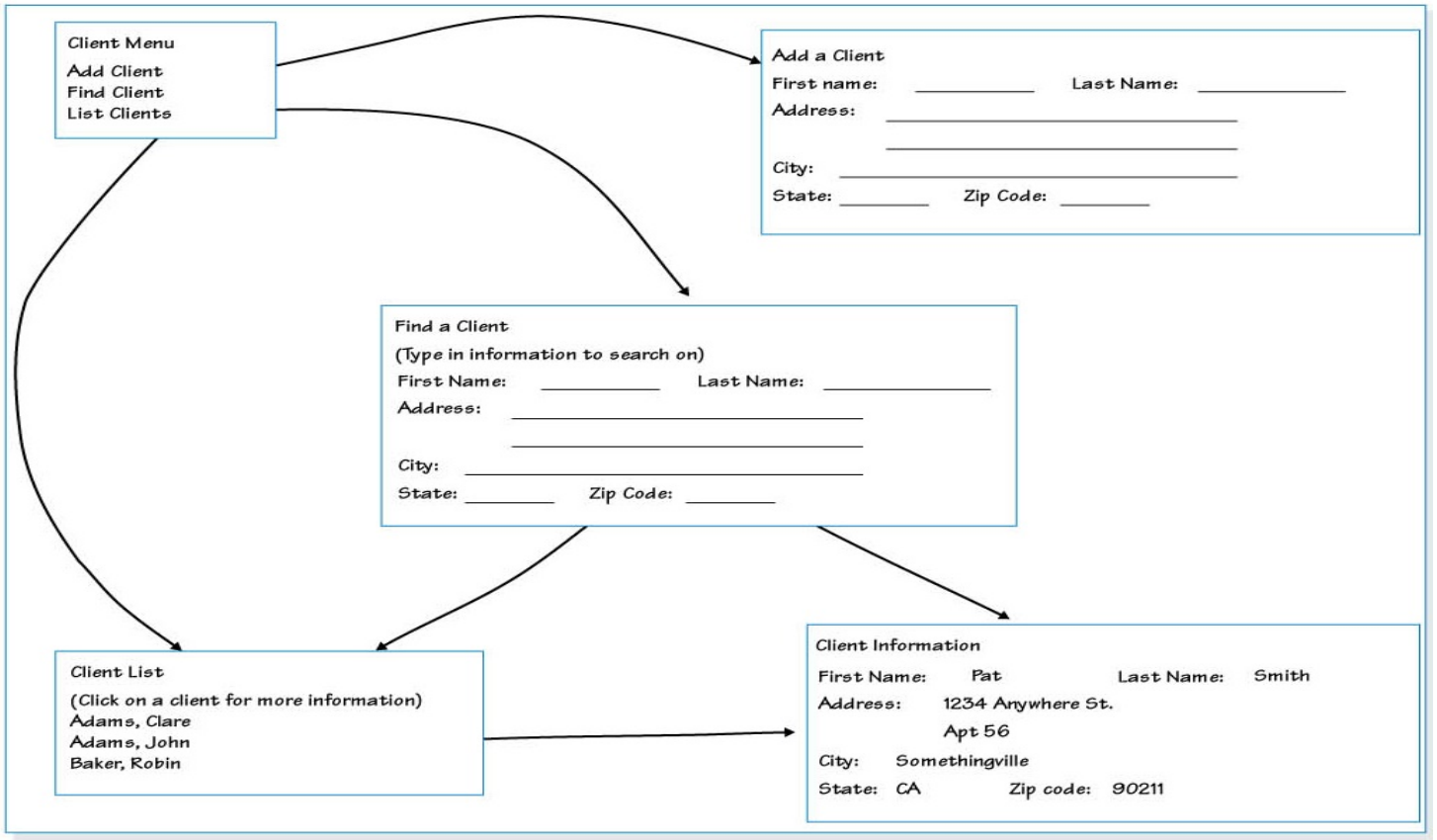
- The template specifies the names that the interface will use for the major ***interface objects***, the fundamental building blocks of the system.
- The template gives names to the most commonly used *interface actions*.
- The interface objects and actions, and also their status, may be represented by ***interface icons***.

Interface Design Prototyping

- An interface design prototype is a mock-up or a simulation of a computer screen, form, or report.
- Common approaches to interface design prototyping:
  - Storyboards
  - HTML prototypes
  - Language prototypes.

Storyboard

- The ***storyboard*** shows hand-drawn pictures of screens.

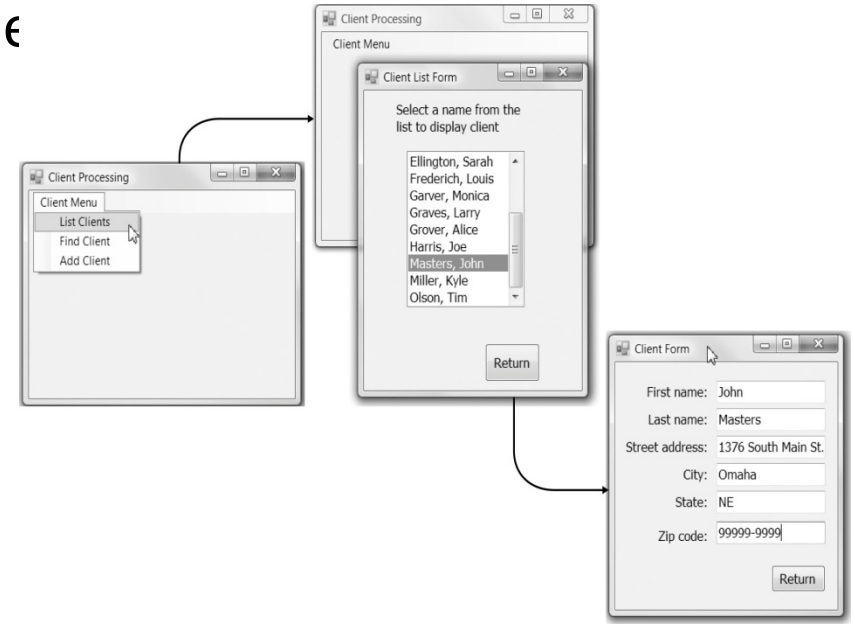


HTML Prototype

- An ***HTML prototype*** is built with the use of Web pages created in HTML (hypertext mark-up language).
- The designer uses HTML to create a series of Web pages that show the fundamental parts of the system.

Language Prototype

- A language prototype is an interface design prototype built in the actual language or by the actual that will be used to build the
- An example of language prototype.



Interface Evaluation

- The objective of interface evaluation is to understand how to improve the interface design.
- There are four common approaches to interface evaluation.

1. ***Heuristic evaluation -*** Compare the interface to a checklist of design principles.
2. ***Walk-through evaluation -*** It is a meeting conducted with the users to walk through the interface.
3. ***Interactive evaluation -*** Users try out the interface.
4. ***Formal usability testing -*** It is a formal testing process to understand how usable the interface is.

Navigation Design

- **Basic Principles**

- Analysts usually must assume that users have not read the manual, have not attended training, and do not have external help readily at hand.
- All controls should be clear and understandable and placed in an intuitive location on the screen.

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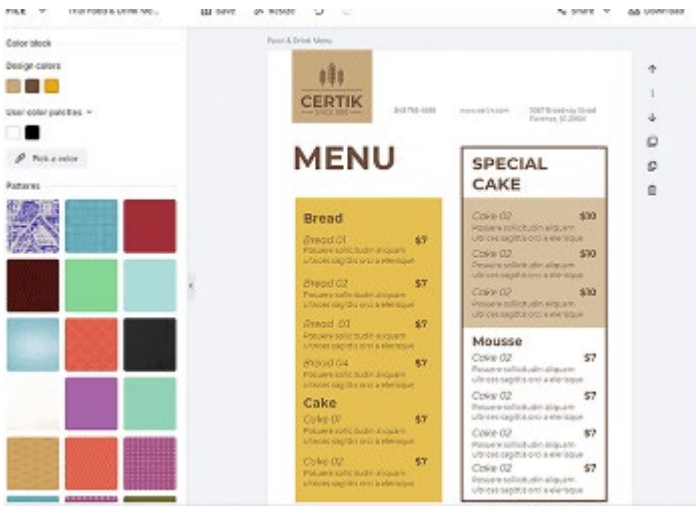
- **Prevent Mistakes** - The first of principle of designing navigation control is to prevent users from making mistakes.
  - Labeling commands appropriately and limiting choices.
  - Confirming with the user that the actions are difficult or impossible to undo.
- **Simplify Recovery from Mistakes** – making "undo" buttons whenever possible.
- **Use Consistent Grammar Order** – Windows application uses an *object-action* grammar order.

Menu Tips

- The most common type of navigation system today is the menu. A menu presents the user with a list of choices, each of which can be selected
- Menus should be designed with care, because the submenus behind a main menu are hidden from users until they click on the menu item.



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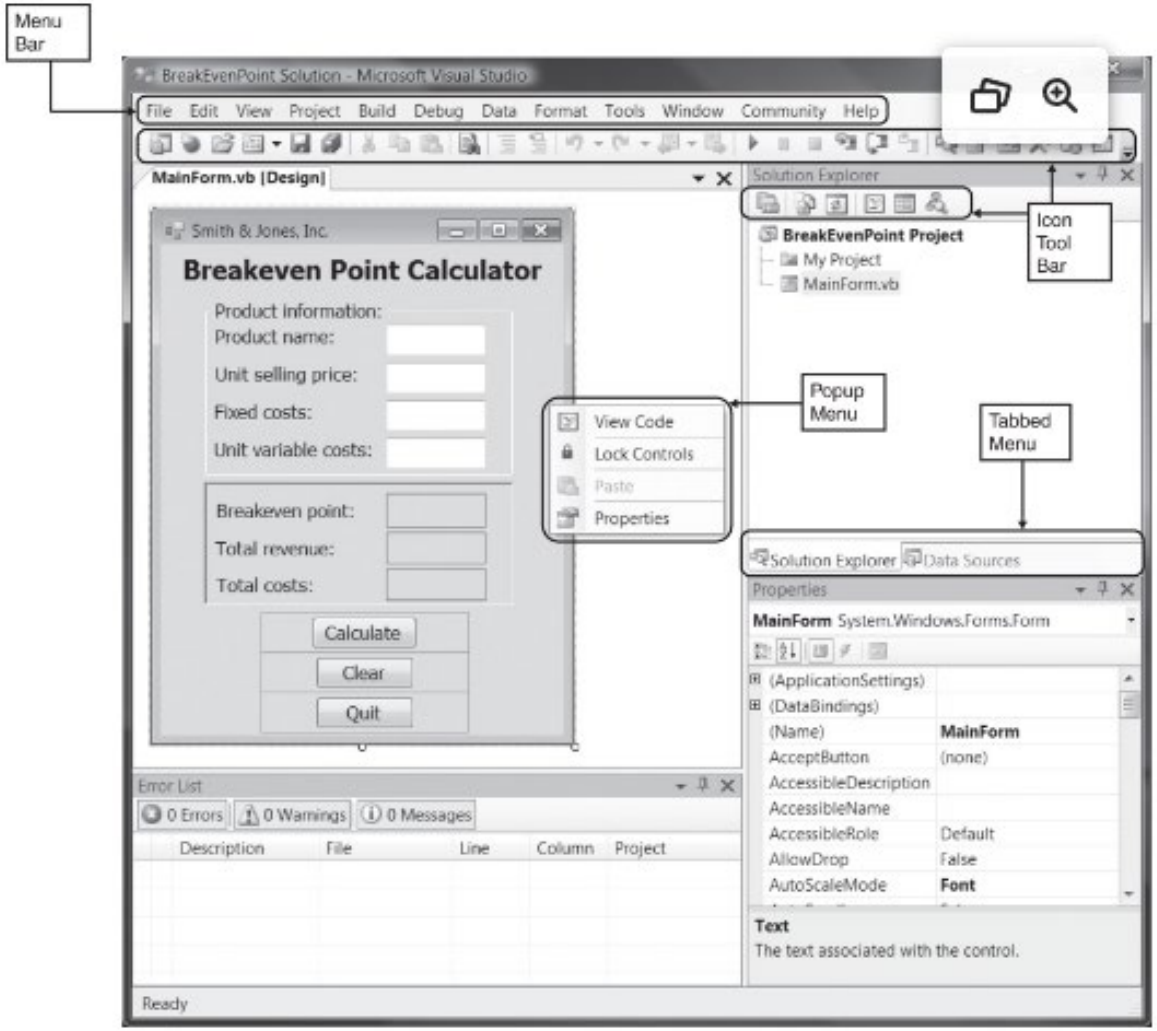

FIGURE 8-15 Common types of menus—Part A.

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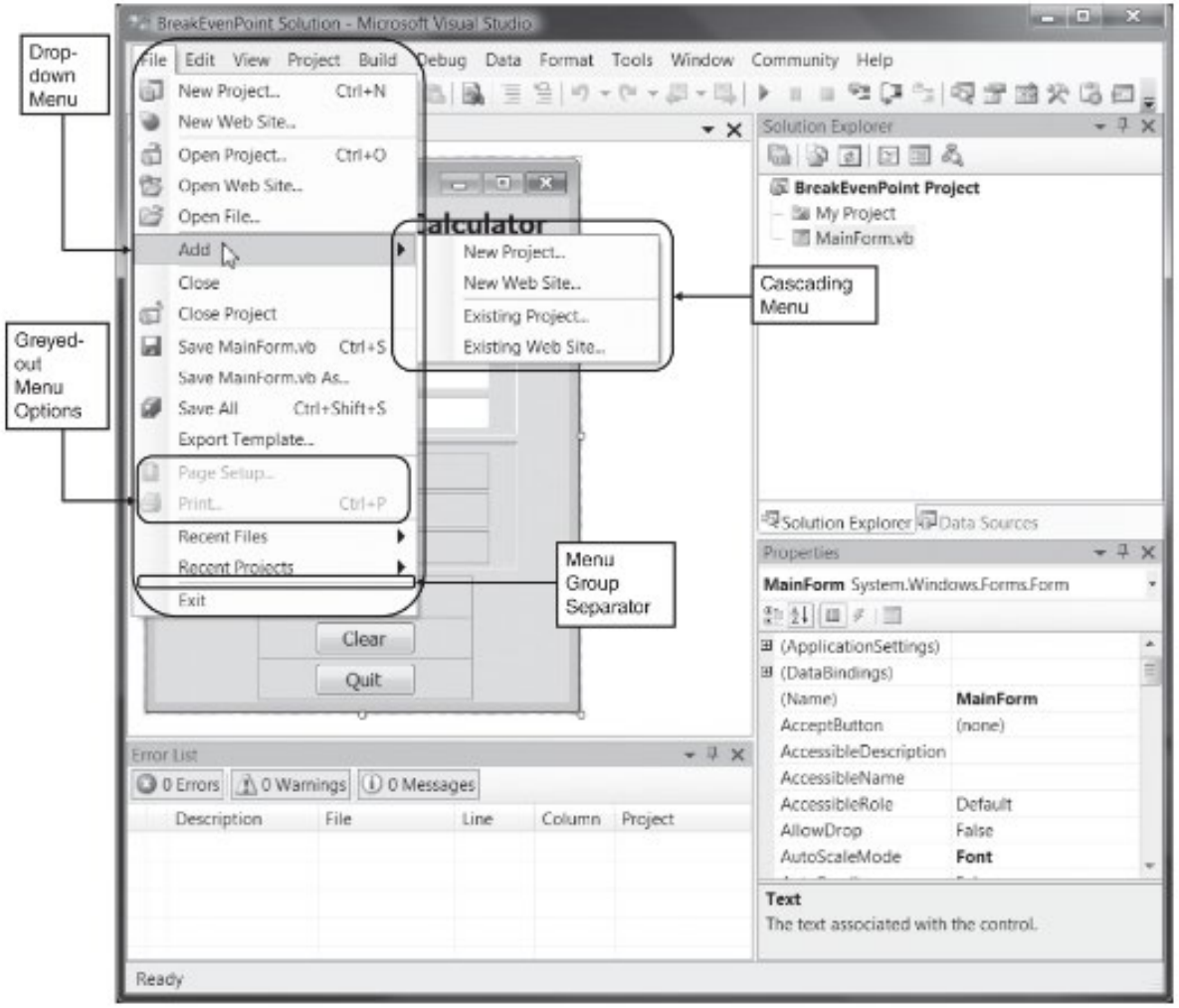


FIGURE 8-15 Common types of menus—Part B.

FIGURE 8-16:
Types of menus.

Type of Menu	When to Use	Notes
Menu Bar List of commands at the top of the screen. Always on screen.	Main menu for system	• Use the same organization as the operating system and other packages (e.g., File, Edit, View). • Menu items are always one word, never two. • Menu items lead to other menus, rather than performing action. • Never allow users to select actions they cannot perform. (Instead, use grayed-out items).
Drop-down Menu Menu that drops down immediately below another menu. Disappears after one use.	Second-level menu, often from menu bar	• Menu items are often multiple words. • Avoid abbreviations • Menu items perform action or lead to another cascading drop-down menu, pop-up menu, or tab menu.
Hyperlink Menu A set of items arranged as a menu, usually along one edge of the screen.	Main menu for Web-based system	• Most users are familiar with hyperlink menus on the left edge of the screen, although they can be placed along any edge. • Menu items are usually only one or two words.
Embedded Hyperlinks A set of items embedded and underlined in text.	As a link to ancillary, optional information	• Used sparingly to provide additional information, because they can complicate navigation. • Usually open a new window that is closed once the action is complete so that the user can return to the original use scenario.
Pop-up Menu Menu that pops up and floats over the screen. Disappears after one use	As a shortcut to commands for experienced users	• Often (not always) invoked by a right click in Windows-based systems. • Menu choices vary depending on pointer position. • Often overlooked by novice users, so usually should duplicate functionality provided in other menus.
Tab Menu Multipage menu with one tab for each page that pops up and floats over the screen. Remains on screen until closed.	When user needs to change several settings or perform several related commands	• Menu items should be short to fit on the tab label. • Avoid more than one row of tabs because clicking on a tab to open it can change the order of the tabs, and in virtually no other case does selecting from a menu rearrange the menu itself.
Tool Bar Menu of buttons (often with icons) that remains on the screen until closed.	As a shortcut to commands for experienced users	• All buttons on the same tool bar should be the same size. • If the labels vary dramatically in size, then use two different sizes (small and large). • Buttons with icons should have a tool tip—an area that displays a text phrase explaining the button when the user pauses the pointer over it.
Image Map Graphical image in which certain areas are linked to actions or other menus.	Only when the graphical image adds meaning to the menu	• Image should convey meaning to show which parts perform an action when clicked. • Tool tips can be helpful.

Messages

- Messages are the way in which the system responds to a user and informs the user of the status of the interaction.
- Messages should be clear, concise, and complete.
- All messages should be grammatically correct and free of jargon and abbreviations (unless they are the users' ones).
- Avoid negatives and humor.

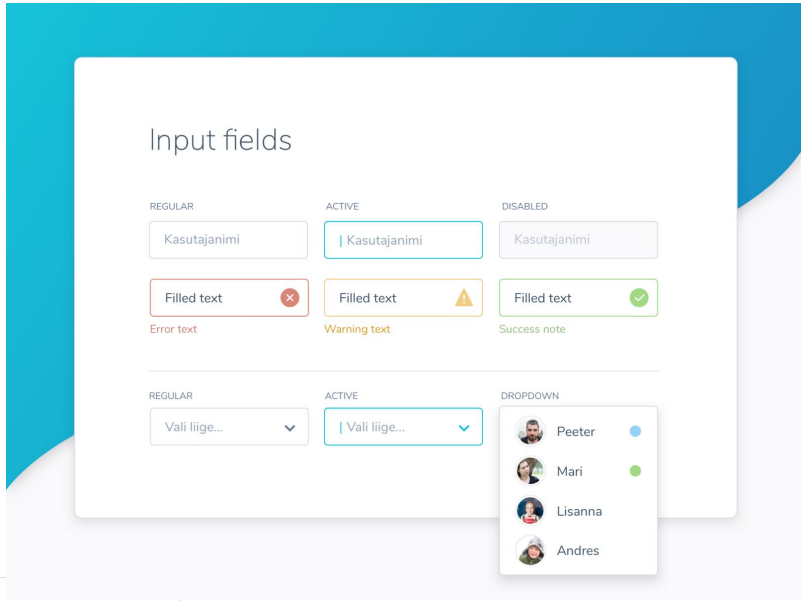
(cont'd)

Type of Messages	When to Use	Notes
Error message Informs the user that he or she has attempted to do something to which the system cannot respond.	When user does something that is not permitted or not possible	Always explain the reason and suggest corrective action. Traditionally, error messages have been accompanied by a beep, but many applications now omit it or permit users to remove it.
Confirmation message Asks the user to confirm that he or she really wants to perform the action selected.	When user selects a potentially dangerous choice, such as deleting a file	Always explain the cause and suggest possible action. Often include several choices other than "OK" and "cancel."
Acknowledgment message Informs the user that the system has accomplished what it was asked to do.	Seldom or never; users quickly become annoyed with all the unnecessary mouse clicks	Acknowledgment messages are typically included because novice users often like to be reassured that an action has taken place. The best approach is to provide acknowledgment information without a separate message on which the user must click. For example, if the user is viewing items in a list and adds one, then the updated list on the screen showing the added item is sufficient acknowledgment.
Delay message Informs the user that the computer system is working properly.	When an activity takes more than seven seconds	This message should permit the user to cancel the operation in case he or she does not want to wait for its completion. The message should provide some indication of how long the delay may last.
Help message Provides additional information about the system and its components.	In all systems	Help information is organized by table of contents and/or keyword search. Context-sensitive help provides information that is dependent on what the user was doing when help was requested. Help messages and online documentation are discussed in Chapter 11.

FIGURE 8-17 Types of messages.

Input Design

- Input mechanisms facilitate the entry of data into the computer system.
- Input design means designing the screen used to enter information and forms on which the users write and type information.



Basic Principles

- The goal of input design is to capture accurate information for the system simply and easily.

Use Online and Batch Processing Appropriately

- There are two general formats for entering inputs into a computer system: online processing and batch processing.
- **Online processing**: each input item is entered into the system immediately.
- **Batch processing**: all the inputs collected over some period are gathered together and entered into the system at one time in a batch.
- Batch processing simplifies data communications and cuts communications costs.

Capture Data at the Source

- The most important principle of input design is to capture the data in an electronic format at the original source.
- It reduces duplication work, reduces, processing time, decreases the cost, decreases the probability of error.

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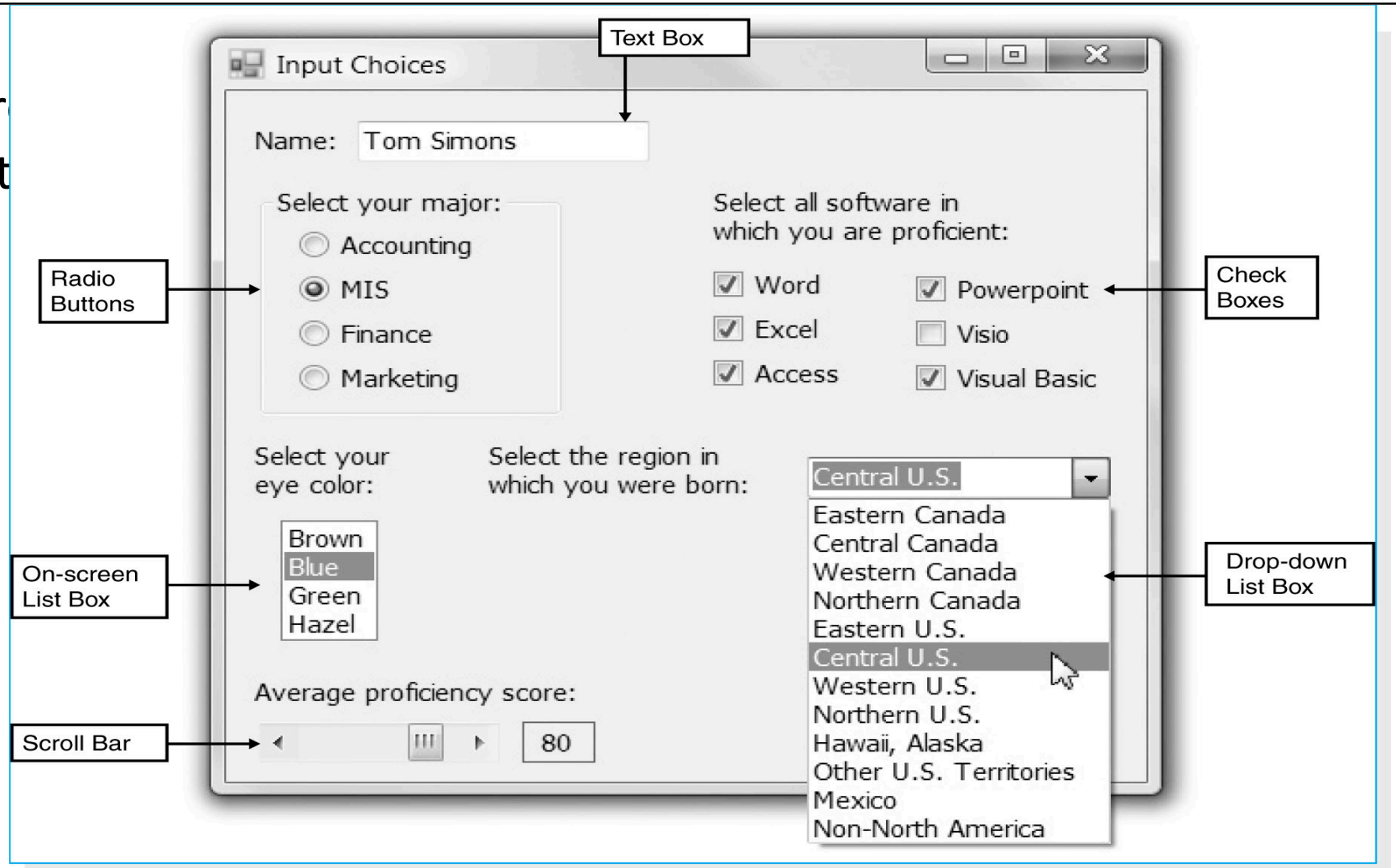
- ***Source data automation*** refers to using special hardware devices to automatically capture data without requiring anyone to type it.
- Source data automation technologies:
  - bar code readers
  - optical character recognition
  - magnetic stripe readers
  - smart cards
  - RFID (radio frequency identification) tags
  - the Web.

Minimize Keystrokes

- Keystrokes cost time and money.
- The system should never ask for information that can be obtained in another way (e.g., by retrieving it from a database).
- The system should not require a user to type information that can be selected from a list.
- The frequent values should be used as the ***default value*** for the data.

Types of Inputs

- There ar many different



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Type of Box	When to Use	Notes
Check box Presents a complete list of choices, each with a square box in front	When several items can be selected from a list of items	Check boxes are not mutually exclusive. Do not use negatives for box labels. Check box labels should be placed in some logical order, such as that defined by the business process, or failing that, alphabetically or most commonly used first. Use no more than ten check boxes for any particular set of options. If you need more boxes, group them into subcategories.
Radio button Presents a complete list of mutually exclusive choices, each with a circle in front	When only one item can be selected from a set of mutually exclusive items	Use no more than six radio buttons in any one list; if you need more, use a drop-down list box. If there are only two options, one check box is usually preferred to two radio buttons, unless the options are not clear. Avoid placing radio buttons close to check boxes to prevent confusion between different selection lists.
On-screen list box Presents a list of choices in a box	Seldom or never—only if there is insufficient room for check boxes or radio buttons	This type of box can permit only one item to be selected (in which case it is an ugly version of radio buttons). This type of box can also permit many items to be selected (in which case it is an ugly version of check boxes), but users often fail to realize they can choose multiple items. This type of box permits the list of items to be scrolled, thus reducing the amount of screen space needed.
Drop-down list box Displays selected item in one-line box that opens to reveal list of choices	When there is insufficient room to display all choices	This type of box acts like radio buttons but is more compact. This type of box hides choices from users until it is opened, which can decrease ease of use; conversely, because it shelters novice users from seldom-used choices, it can improve ease of use. This type of box simplifies design if the number of choices is unclear, because it takes only one line when closed.
Combo box A special type of drop-down list box that permits user to type as well as scroll the list	Shortcut for experienced users	This type of box acts like drop-down list but is faster for experienced users when the list of items is long.
Slider Graphic scale with a sliding pointer to select a number	Entering an approximate numeric value from a large continuous scale	The slider makes it difficult for the user to select a precise number. Some sliders also include a number box to enable the user to enter a specific number.

Input Validation

- All data entered into the system must be validated in order to ensure accuracy.
- ***Input validation*** (also called ***edit checks***) can take many forms

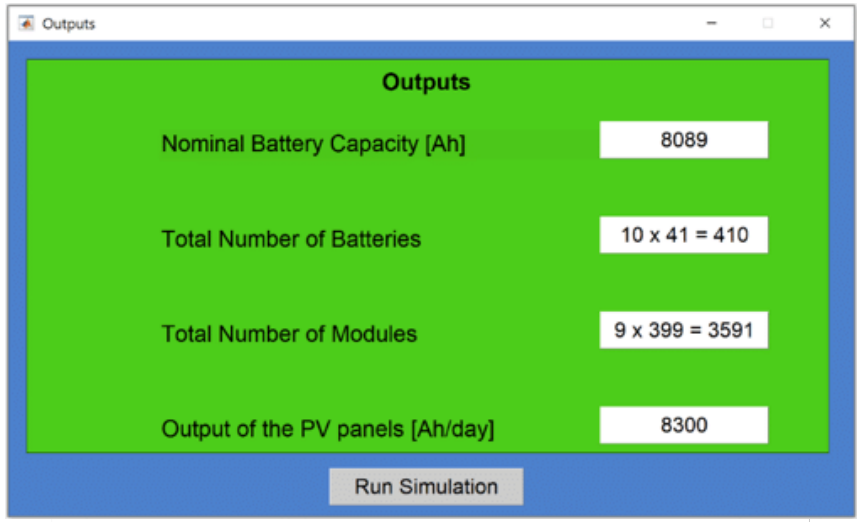
(cont'd)

- There are

Type of Validation	When to Use	Notes
Completeness check Ensures all required data have been entered	When several fields must be entered before the form can be processed	If required information is missing, the form is returned to the user unprocessed.
Format check Ensures data are of the right type (e.g., numeric) and in the right format (e.g., month, day, year)	When fields are numeric or contain coded data	Ideally, numeric fields should not permit users to type text data, but if this is not possible, the entered data must be checked to ensure it is numeric. Some fields use special codes or formats (e.g., license plates with three letters and three numbers) that must be checked.
Range check Ensures numeric data are within correct minimum and maximum values	With all numeric data, if possible	A range check permits only numbers between correct values. Such a system can also be used to screen data for "reasonableness"—e.g., rejecting birthdates prior to 1880 because people do not live to be a great deal over 100 years old (most likely, *1980* was intended).
Check digit check Check digits are added to numeric codes	When numeric codes are used	Check digits are numbers added to a code as a way of enabling the system to quickly validate correctness. For example, U.S. Social Security Numbers and Canadian Social Insurance Numbers assign only eight of the nine digits in the number. The ninth number—the check digit—is calculated using a mathematical formula from the first eight numbers. When the identification number is typed into a computer system, the system uses the formula and compares the result with the check digit. If the numbers don't match, then an error has occurred.
Consistency checks Ensure combinations of data are valid	When data are related	Data fields are often related. For example, someone's birth year should precede the year in which he or she was married. Although it is impossible for the system to know which data are incorrect, it can report the error to the user for correction.
Database checks Compare data against a database (or file) to ensure they are correct	When data are available to be checked	Data are compared against information in a database (or file) to ensure they are correct. For example, before an identification number is accepted, the database is queried to ensure that the number is valid. Because database checks are more "expensive" than the other types of checks (they require the system to do more work), most systems perform the other checks first and perform database checks only after the data have passed the previous checks.

Output design

- Outputs are the reports that the system produces, whether on the screen, on paper, or in other media, such as the Web.
- Outputs are the most visible part of any system.

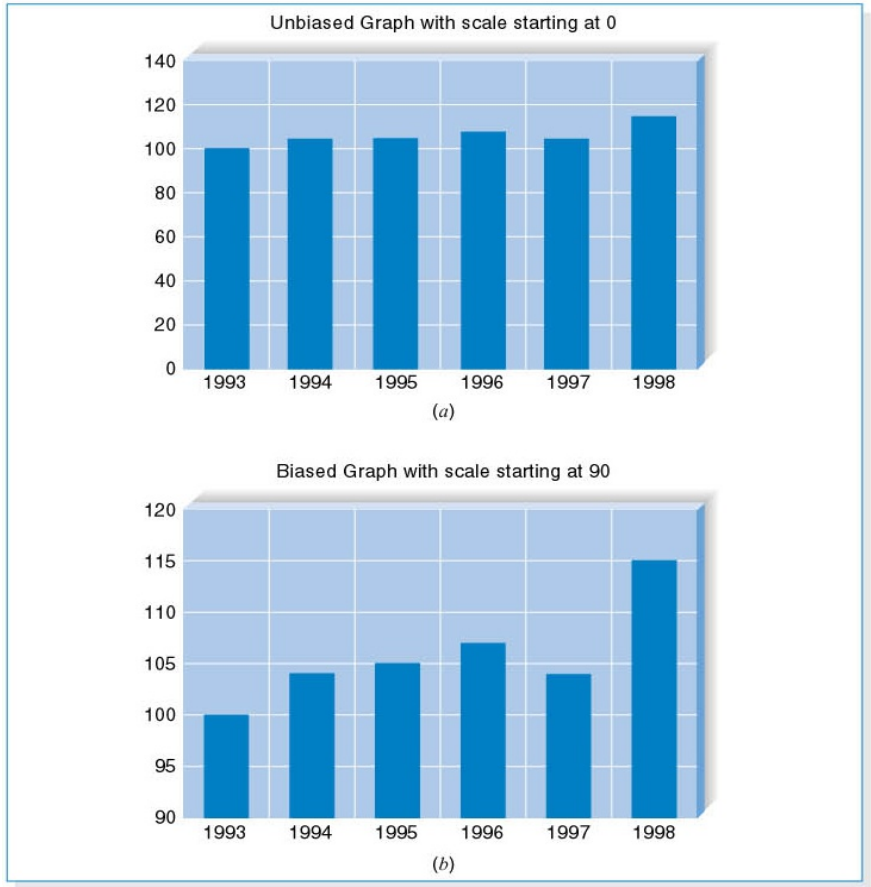


Basic Principles

- The goal of the output mechanism is to present information to users so that they can accurately understand it with the least effort.
- **Understand report usage** – the first principle in designing reports is to understand how they are used.
- **Manage information load** – the goal of a well-designed report is to provide all needed information without information overload.
- **Minimize bias** – no analyst sets out to design a biased report.

(cont'd)

- Example of bias: Bias in graphs.



Types of Outputs

Type of Reports	When to Use	Notes
Detail report Lists detailed information about all the items requested	When user needs full information about the items	This report is usually produced only in response to a query about items matching some criteria. This report is usually read cover to cover to aid understanding of one or more items in depth.
Summary report Lists summary information about all items	When user needs brief information on many items	This report is usually produced only in response to a query about items matching some criteria, but it can be a complete database. This report is usually read for the purpose of comparing several items to each other. The order in which items are sorted is important.
Turnaround document Outputs that "turn around" and become inputs	When a user (often a customer) needs to return an output to be processed	Turnaround documents are a special type of report that are both outputs and inputs. For example, most bills sent to consumers (e.g., credit card bills) provide information about the total amount owed and also contain a form that consumers fill in and return with payment.
Graphs Charts used in addition to and instead of tables of numbers	When users need to compare data among several items	Well-done graphs help users compare two or more items or understand how one has changed over time. Graphs are poor at helping users recognize precise numeric values and should be replaced by or combined with tables when precision is important. Bar charts tend to be better than tables of numbers or other types of charts when it comes to comparing values between items (but avoid three-dimensional charts that make comparisons difficult). Line charts make it easier to compare values over time, whereas scatter charts make it easier to find clusters or unusual data. Pie charts show proportions or the relative shares of a whole.

Media

- The two dominant media of reports are paper and electronic.
- Paper is the more traditional medium and is relatively permanent, easy to use, highly portable, and accessible in most situations.
- However, paper reports are expensive.
- Many organizations are moving to electronic production of reports.

SUMMARY

- User interface design principles

- Layout, content awareness, aesthetics, user experience, consistency, minimize user effort.

- The user interface design process

- Use scenario development, interface structure design, interface standards design, interface design prototyping, and interface evaluation.

(cont'd)

- Navigation design

  - The fundamental goal of navigation design is to make the system as simple to use as possible.

- Input design

  - The goal of input design is to simply and easily capture accurate information for the system.

- Output design

  - The goal of the output design is to present information to users so that they can accurately understand it with the least effort.